

Homework 1

- This is a modified version of a recent final exam given in ST 501.
- You are allowed to discuss answers and generally collaborate on assignments in this class.
- However, you are not allowed to copy answers/explanations. You should be writing things in your own words and writing out derivations yourself.
- If you are ever questioning whether a collaboration is allowed, please just ask!
- **Although there are points associated with each problem, only selected problems will be graded for correctness, the rest will be graded for completeness (and not worth the points as noted).** The points are just to show you how I would allocate this on an exam.

1. Briefly answer the following:

- (a) What does the term distribution mean in non-technical terms? (3 pts)

The pattern and frequency with which we observe a random variable (or data).

- (b) In non-technical terms, what is meant by the phrase 'sampling distribution of the sample mean'? (2 pts)

The pattern and frequency in which we observe a sample mean.

- (c) Practically speaking, what does the Central Limit Theorem tell us (be specific about assumptions required and clear about what statistics it applies to!) and why is it so useful? (5 pts)

The CLT tells us that sample averages and sample sums from a random sample will roughly follow a normal distribution in large samples, provided the population has finite variance. This is very useful for finding approximate probabilities that can be used to make inferences when it might otherwise be difficult to derive exact results.

- (d) In regards to statistical inference, why do we often make the assumption that a sample is 'iid'? (3 pts)

By assuming a sample is iid, we don't need to specify the joint behavior of the random variables together, we can just specify the marginal behavior and obtain the joint from the product of the marginals. By assuming identically distributed, we are really just assuming our data comes from the same population.

Again, please note: **Show all of your work and explain non-algebra steps** to receive full credit.

2. Suppose that $Y_i \stackrel{iid}{\sim} \exp(\lambda)$.

(a) Derive the mean and variance of $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$. (3 and 5 pts, respectively)

$$\begin{aligned} E(\bar{Y}) &= E\left(\frac{1}{n} \sum_{i=1}^n Y_i\right) = \frac{1}{n} \sum_{i=1}^n E(Y_i) && \text{expected value is a linear operator} \\ &= \frac{1}{n} \sum_{i=1}^n \frac{1}{\lambda} && \text{identically distributed} \\ &= \frac{1}{\lambda} && \text{Note: same as the population mean} \end{aligned}$$

$$\begin{aligned} Var(\bar{Y}) &= Var\left(\frac{1}{n} \sum_{i=1}^n Y_i\right) \\ &= \frac{1}{n^2} \sum_{i=1}^n Var(Y_i) + 0 && \text{square constants, independence implies covariance terms are 0} \\ &= \frac{1}{n^2} \sum_{i=1}^n \frac{1}{\lambda^2} && \text{identically distributed} \\ &= \frac{1}{n\lambda^2} && \text{Note: population variance divided by } n \end{aligned}$$

(b) Give a detailed argument (state any theorems and assumptions) to show that a reasonable large-sample distribution for $\bar{Y}$ is a Normal distribution (and of course give the mean and variance). (6 pts)

The CLT can be applied to this sample mean because we have an iid sample from a population with finite variance. The CLT gives that

$$\bar{Y} \stackrel{\bullet}{\sim} N(\mu, \sigma^2/n) = N(1/\lambda, 1/(\lambda^2 n))$$

(c) Give an equation in terms of a standard normal random variable's CDF, n , and λ that will approximate $P(\bar{Y} > 20)$. (3 pts)

$$P(\bar{Y} > 20) = P\left(\frac{\bar{Y} - 1/\lambda}{1/(\lambda\sqrt{n})} > \frac{20 - 1/\lambda}{1/(\lambda\sqrt{n})}\right) \approx 1 - P\left(Z \leq \frac{20 - 1/\lambda}{1/(\lambda\sqrt{n})}\right)$$

- (d) (This question is part of the previous problem.) Find a large-sample distribution for the $\ln(\bar{Y})$ (don't derive the exact distribution here, give an approximating distribution). (6 pts)

Define $W = \ln(\bar{Y})$. Using the Delta Method we can approximate the mean and variance W by

$$E(W) \approx g(\mu_{\bar{Y}}) = \ln(1/\lambda)$$

$$Var(W) \approx [g'(\mu_{\bar{Y}})]^2 \sigma_{\bar{Y}}^2 = \left[\frac{1}{1/\lambda} \right]^2 \frac{1}{n\lambda^2} = 1/n$$

Using delta method normality, a large-sample distribution for W is then

$$W \overset{\bullet}{\sim} N(\ln(1/\lambda), 1/n)$$

- (e) (This question is part of the previous problem.) Consider $S^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2$. Prove that $S^2 \xrightarrow{p} 1/\lambda^2$. Again, be sure to state theorems and results you use. (8 pts)

We can't apply the LLN because we don't have independence of the terms in our sum. However, we can expand out the square, distribute the sum, and utilize results on those terms.

$$\begin{aligned} S^2 &= \frac{1}{n} \left[\sum_{i=1}^n Y_i^2 - 2\bar{Y} \sum_{i=1}^n Y_i + n(\bar{Y})^2 \right] \\ &= \frac{1}{n} \sum_{i=1}^n Y_i^2 - 2(\bar{Y})^2 + \bar{Y}^2 \\ &= \frac{1}{n} \sum_{i=1}^n Y_i^2 - (\bar{Y})^2 \end{aligned}$$

Now we can apply the LLN to the first term (Y_i^2 being our RV), noting that

$$E(Y_i^2) = \text{Var}(Y_i) + (E(Y_i))^2 = 1/\lambda^2 + (1/\lambda)^2 = 2/\lambda^2$$

For the second term, by the LLN we know that $\bar{Y}$ converges to $1/\lambda$.

By the continuity theorem we have that $(\bar{Y})^2$ converges to $1/\lambda^2$.

Using 'standard limiting results' we can now say

$$S^2 = \frac{1}{n} \sum_{i=1}^n Y_i^2 - (\bar{Y})^2 \xrightarrow{p} 2/\lambda^2 - 1/\lambda^2 = 1/\lambda^2 = \text{Var}(Y)$$

(f) (This question is part of the previous problem.) Give an argument as to why

$$Z = \frac{\bar{Y} - (1/\lambda)}{S/\sqrt{n}}$$

converges in distribution to a Normal distribution. Again, be sure to state theorems and results you use. (6 pts)

We know from the CLT that

$$\frac{\bar{Y} - 1/\lambda}{\sqrt{\frac{1}{n\lambda^2}}} \xrightarrow{d} Z \sim N(0,1)$$

By the previous problem, we have that

$$S^2 \xrightarrow{p} 1/\lambda^2$$

Using continuity we know that

$$\sqrt{\frac{\frac{1}{\lambda^2}}{S^2}} \xrightarrow{p} 1$$

Now we can apply Slutsky's theorem to say that

$$\frac{\bar{Y} - 1/\lambda}{\sqrt{\frac{1}{n\lambda^2}}} \sqrt{\frac{\frac{1}{\lambda^2}}{S^2}} = \frac{\bar{Y} - (1/\lambda)}{S/\sqrt{n}} \xrightarrow{d} Z(1) = Z \sim N(0,1)$$

3. Suppose that $Y \sim \text{geom}(p)$ where Y represents the number of failures until the first success. That is, we have support $y=0,1,2,\dots$ (this differs from the formula sheet version). Information about this version of the geometric is given below.

$$p_Y(y) = p(1-p)^y \quad E(Y) = (1-p)/p \quad \text{Var}(Y) = (1-p)/p^2 \quad m_Y(t) = \frac{p}{1-(1-p)e^t}$$

Please note the geometric sum properties on the formula sheet.

- (a) Derive the MGF of Y given above. (6 pts)

$$m_Y(t) = E(e^{tY}) = \sum_{y=0}^{\infty} e^{ty}(1-p)^y p = p e^t \sum_{y=0}^{\infty} (e^t(1-p))^y = \frac{p e^t}{1 - e^t(1-p)}$$

using the properties of the sum of an infinite geometric series.

- (b) Consider having an integer b . Show that $P(Y > b) = (1-p)^{b+1}$ (6 pts)

$$P(Y > b) = P(Y \geq b+1) = \sum_{y=b+1}^{\infty} (1-p)^y p$$

substituting $x = y - b - 1$ we have

$$\sum_{x=0}^{\infty} (1-p)^{x+b+1} = \sum_{x=0}^{\infty} p(1-p)^{b+1}(1-p)^x = \frac{p(1-p)^{b+1}}{1-(1-p)} = (1-p)^{b+1}$$

- (c) Now suppose we have a random sample, $Y_i \stackrel{iid}{\sim} \text{geom}(p)$. Show that $W = \sum_{i=1}^n Y_i \sim \text{NegativeBinomial}(n, p)$. (8 pts) With this version of the geometric, W represents the number of failures until the n^{th} success and has support $w = 0, 1, 2, \dots$. Information about this version of the negative binomial is given below.

$$p_W(t) = \binom{w+n-1}{w} (1-p)^w p^n \quad E(W) = r(1-p)/p \quad \text{Var}(W) = r(1-p)/p^2 \quad m_W(t) = \left(\frac{p}{1-(1-p)e^t} \right)^n$$

As we have a linear combination of independent RVs, we can try use the MGF to find the distribution.

$$\begin{aligned} m_W(t) &= E\left(e^{t\sum_{i=1}^n Y_i}\right) = E\left(e^{tY_1} e^{tY_2} \dots e^{tY_n}\right) \\ &= E\left(e^{tY_1}\right) E\left(e^{tY_2}\right) \dots E\left(e^{tY_n}\right) && \text{by independence} \\ &= \left(\frac{p}{1-e^t(1-p)} \right)^n && \text{identically distributed} \end{aligned}$$

This is the MGF of a negative binomial with parameters n and p

- (d) Suppose that $n \rightarrow \infty$ and $(1-p) \rightarrow 0$ in such a way that $n(1-p) \rightarrow \lambda$. Show that $W \xrightarrow{d} X \sim Poi(\lambda)$. Recall our common limit trick, which appears on the formula sheet. (8 pts)
We'll use convergence of MGFs to show this

$$\begin{aligned}
 m_W(t) &= \left(\frac{p}{1-e^t(1-p)} \right)^n \\
 &= \left(1 - 1 + \frac{p}{1-e^t(1-p)} \right)^n \\
 &= \left(1 + \frac{-1+e^t(1-p)+p}{1-e^t(1-p)} \right)^n \\
 &= \left(1 + \frac{(1-p)(e^t-1)}{1-e^t(1-p)} \right)^n \\
 &= \left(1 + \frac{\left(\frac{n(1-p)(e^t-1)}{1-e^t(1-p)} \right)}{n} \right)^n
 \end{aligned}$$

Now we can apply the common limiting result

$$(1 + a_n/n)^n \rightarrow e^a$$

where $a_n \rightarrow a$. Noting that

$$\frac{n(1-p)(e^t-1)}{1-e^t(1-p)} \rightarrow \frac{\lambda(e^t-1)}{1-0} = \lambda(e^t-1)$$

then

$$m_W(t) \rightarrow e^{\lambda(e^t-1)}$$

which is the MGF of a $Poi(\lambda)$ RV.

4. Let $Y \sim \text{Beta}(\alpha, \beta)$.

(a) Derive the result $E(Y^r) = \frac{\Gamma(\alpha+r)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+r+\beta)}$. Assume $r > -\alpha$. (6 pts)

$$E(Y^r) = \int_0^1 y^r \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} y^{\alpha-1} (1-y)^{\beta-1} dy = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 y^{\alpha+r-1} (1-y)^{\beta-1} dy$$

We can recognize the integral as the kernel of a Beta distribution with parameters $\alpha+r$ and β (which is ok since the assumption is that $r > -\alpha$ which implies $\alpha+r > 0$).

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(\alpha+r)\Gamma(\beta)}{\Gamma(\alpha+r+\beta)} = \frac{\Gamma(\alpha+r)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+r+\beta)}$$

(b) Use the above result to find the variance of Y . (4 pts)

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2 = \frac{\Gamma(\alpha+2)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+2+\beta)} - \left[\frac{\Gamma(\alpha+1)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+1+\beta)} \right]^2$$

Using properties of the $\Gamma(\cdot)$ function we can simplify this a bit:

$$= \frac{(\alpha+1)(\alpha)}{(\alpha+\beta+1)(\alpha+\beta)} - \left[\frac{\alpha}{\alpha+\beta} \right]^2 = \frac{(\alpha+1)(\alpha+\beta)\alpha - \alpha^2(\alpha+\beta+1)}{(\alpha+\beta+1)(\alpha+\beta)^2} = \frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2}$$

- (c) **Two things:** Provide an equation that uses the PDF (no need to solve the equation) that corresponds to $P(0.2 < Y \leq 0.4)$ and draw a detailed picture of the PDF corresponding to this probability (pick any reasonable shape for the distribution of a Beta when drawing your picture). (6 pts)

$$P(0.2 < Y \leq 0.4) = \int_{0.2}^{0.4} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} y^{\alpha-1} (1-y)^{\beta-1} dy$$

Visual omitted.

- (d) **Two things:** Give a detailed equation for how to find the 0.25 quantile of Y's distribution (no need to simplify) using the PDF. Also, draw a detailed picture of this quantile in terms of the PDF. (6 pts)

We want to find $y_{0.25}$ where

$$0.25 = \int_0^{y_{0.25}} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} t^{\alpha-1} (1-t)^{\beta-1} dt$$

Visual omitted.